MARYAM ATHEEQ

Electrical Engineering Student

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WORK EXPERIENCE

System Planning Engineering Co-op Student

Newfoundland Power Inc.

Sep – Dec 2026 St. John's, NL

- Prepared net metering applications by reviewing technical specifications and single-line diagrams, conducting ***load flow*** and ***short-circuit analyses*** in ***CYME*** to assess grid interconnection feasibility
- Analyzed and updated system load forecasts, infeed forecasts, and infeed load splits using ***MS Excel*** and ***PI DataLink*** to evaluate load growth trends and support distribution system planning
- Developed ***Python scripts*** to automate data processing and CYME model updates, streamlining power system modelling workflows and reducing manual effort
- Conducted and updated ***voltage regulation studies*** using ***CYME*** to evaluate feeder voltage profiles and assess distribution system performance
- Collaborated with the Asset Management team on a ***substation switch risk assessment*** project to support equipment replacement strategies

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Electrical Engineering Co-op Student

Cornwall Electric, a FortisOntario Company

Jan - Apr 2026 Cornwall, ON

- Supported distribution system planning activities for transformers, transclosures, underground cables, and vaults using ***MS Excel*** and ***ArcGIS*** to analyze distribution assets
- Developed utility job drawings in ***AutoCAD*** including pole layouts, clearances and joint-use configurations
- Conducted ***field inspections*** to assess pole conditions and support engineering decisions related to asset maintenance and replacement
- Performed utility pole design and analysis using ***SPIDACalc***, evaluating loading conditions and compliance with utility design standards
- Managed engineering work orders and documentation in ***SAP*** to support construction and planning workflows

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Electrical Engineering Intern

Stantec Inc.

May - Aug 2025 St. John's, NL

- Designed and developed ***electrical power and lighting systems*** creating detailed floor plans, lighting layouts, and photometric renderings in compliance with the Canadian Electrical Code (CEC)
- Performed lighting photometric calculations using ***AGi32*** to support lighting analysis, and conducted short-circuit and arc flash studies using ***ETAP*** with on-site data collection
- Created single-line diagrams using ***AutoCAD*** and developed ***Revit 3D*** models to support building system layouts

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Technical Project Lead

Suncor Energy Fluvarium

Sep - Dec 2024 St. John's, NL

- Designed and developed small-scale boats using ***SolidWorks*** and fluid mechanics principles, integrating DC motors, Electronic Speed Controllers (ESCs), and wireless communication modules for remote operation
- Led ***project management*** activities, coordinating project timelines, budgets, and sponsor deliverables using ***MS Excel*** and ***MS PowerPoint***

EDUCATION

Bachelor of Electrical Engineering, Co-op Program

Memorial University of Newfoundland

Sep 2022 – Present St. John's, NL

Expected May 2027

AWARDS & SCHOLARSHIPS

- Guinness World Record, Largest Mosaic
- H. John McDonald Memorial Bursary
- Verafin Inc. Engineering Scholarship
- PEGNL Bursary
- International Entrance Scholarship
- Engineering High Achiever Scholarship

PERSONAL PROJECTS

Security Light System

Designed an **Arduino-based** security light system with motion and light sensors, providing automated and energy-efficient lighting solutions

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Prosthetic Hand

Designed and prototyped a 3D-printed prosthetic hand by integrating micro-servo motors and programming a **Raspberry Pi** to control finger motion

TECHNICAL SKILLS

MS Office | CYME | SPIDAcalc | ArcGIS | AGi32 | AutoCAD | Revit | Power Systems | Asset Management | LTSpice | Simulink | KiCad | Circuit Design | Lab Equipment | Project Management | SAP | Pi DataLink | MATLAB | Python | C++ | Arduino UNO | Raspberry Pi | SolidWorks | Onshape | 3D Printing | HTML | CSS | JavaScript

INTERESTS

Volunteering • Hiking • Cooking • Travelling • Photography • Cats